

SPRING 2025
CSE423 QUIZ
Time: 30 Minutes

A

Name: _____ ID: _____ Section: _____

1. You have an axis-aligned rectangular window which has width of 80 pixels, height of 50 pixels and top left corner situated at (-40, 20). Find out the outcode of the following coordinates: (1x5=5)

- a. (0, 0)
- b. (50, 20)
- c. (20, 50)
- d. (-50, -30)
- e. (50, -40)

3	2	1	0
Above	Below	Right	Left

1001	1000	1010
0001	0000 Window	0010
0101	0100	0110

2. In which case Cyrus-Beck Algorithm fails to generate the correct answer? Why does it fail? Provide an example with proper justification. 3
3. Formulate the parametric equation for a line segment going from (-2, 3) to (4, -5) and determine the middle point of the line segment (Hint: use $t = 0.5$) 3
4. Given (0, 0) and (40, -80) are the corners of the clip rectangle. The endpoints of a line are given as (-10, 5) and (50, -110). Identify whether the lines are Accepted, Rejected, or Partially accepted. If partially accepted, find the endpoints of the accepted line segment. 9

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B

Name: _____ ID: _____ Section: _____

1. You have an axis-aligned rectangular window which has width of 80 pixels, height of 50 pixels and top left corner situated at (-20, 40). Find out the outcode of the following coordinates: (1x5=5)

- a. (0, 0)
- b. (-20, 50)
- c. (50, -20)
- d. (-30, 50)
- e. (40, -50)

3	2	1	0
Above	Below	Right	Left

1001	1000	1010
0001	0000 Window	0010
0101	0100	0110

2. In which case Cyrus-Beck Algorithm fails to generate the correct answer? Why does it fail? Provide an example with proper justification. 3
3. Formulate the parametric equation for a line segment going from (-3, 2) to (5, -4) and determine the middle point of the line segment (Hint: use $t = 0.5$) 3
4. Given (0, 0) and (80, -40) are the corners of the clip rectangle. The endpoints of a line are given as (-5, 10) and (110, -50). Identify whether the lines are Accepted, Rejected, or Partially accepted. If partially accepted, find the endpoints of the accepted line segment. 9